

GANGHUA WANG

Email: ganghua@uchicago.edu

Address: Data Science Institute, 5460 S University Ave, Room 202, Chicago, IL, 60615

Research Interests: Trustworthy AI, Machine learning theory and applications, Large generative models

PROFESSIONAL EXPERIENCE

University of Arizona, Department of Mathematics

Assistant Professor

Aug. 2026

Tucson, AZ

University of Chicago, Data Science Institute

Postdoctoral Researcher

Sept. 2024 - present

Chicago, IL

Co-hosted by [Prof. Haifeng Xu](#) and [Prof. Bo Li](#)

EDUCATION

University of Minnesota, Twin Cities

Ph.D. in Statistics

Aug. 2019 - Aug. 2024

Minneapolis, MN

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Peking University

B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019

Beijing, China

PUBLICATIONS

Published

* indicates equal contributions

- [1] **Ganghua Wang**, Yuwei Chen, and Haifeng Xu. “Exponential Convergence of Probabilistic Bisection Algorithm with Noisy Labels”. In: **Proc. ICML** (2026). [\[pdf\]](#).
- [2] **Ganghua Wang**, Yuhong Yang, and Jie Ding. “Model Privacy: A Unified Framework for Understanding Model Stealing Attacks and Defenses”. In: **J. R. Stat. Soc., Ser. B** (2026). [\[pdf\]](#). DOI: [10.1093/jrsssb/qkag059](https://doi.org/10.1093/jrsssb/qkag059).
- [3] Jin Du, Li Chen, Xun Xian, An Luo, Fangqiao Tian, **Ganghua Wang**, Charles Doss, Xiaotong Shen, and Jie Ding. “Ice Cream Doesn’t Cause Drowning: Benchmarking LLMs Against Statistical Pitfalls in Causal Inference”. In: **Proc. ICLR** (2026). [\[pdf\]](#).
- [4] An Luo, Xun Xian, Jin Du, Fangqiao Tian, **Ganghua Wang**, Ming Zhong, Shengchun Zhao, Xuan Bi, Zirui Liu, Jiawei Zhou, et al. “AssistedDS: Benchmarking How External Domain Knowledge Assists LLMs in Automated Data Science”. In: **Proc. EMNLP** (2025). [\[pdf\]](#).
- [5] Xun Xian, **Ganghua Wang**, Xuan Bi, Rui Zhang, Jayanth Srinivasa, Ashish Kundu, Charles Fleming, Mingyi Hong, and Jie Ding. “On the Vulnerability of Applying Retrieval-Augmented Generation within Knowledge-Intensive Application Domains”. In: **Proc. ICML** (2025). [\[pdf\]](#).
- [6] **Ganghua Wang**, Ali Payani, Myungjin Lee, and Ramana Kompella. “Mitigating Group Bias in Federated Learning: Beyond Local Fairness”. In: **Trans. Mach. Learn. Res.** (2024). [\[pdf\]](#).
- [7] **Ganghua Wang**^{*}, Xun Xian^{*}, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Demystifying Poisoning Backdoor Attacks from a Statistical Perspective”. In: **Proc. ICLR** (2024). [\[pdf\]](#).

- [8] Xun Xian, **Ganghua Wang**, Xuan Bi, Jayanth Srinivasa, Ashish Kundu, Mingyi Hong, and Jie Ding. “RAW: A Robust and Agile Plug-and-Play Watermark Framework with Provable Guarantees”. In: **Proc. NeurIPS** (2024). [\[pdf\]](#).
- [9] Enmao Diao*, **Ganghua Wang***, Jie Ding, Yuhong Yang, and Vahid Tarokh. “Pruning deep neural networks from a sparsity perspective”. In: **Proc. ICLR** (2023). [\[pdf\]](#).
- [10] Gen Li, **Ganghua Wang**, and Jie Ding. “Provable Identifiability of Two-Layer ReLU Neural Networks via LASSO Regularization”. In: **IEEE Trans. Inf. Theory** 69.9 (2023), pp. 5921–5935. DOI: [10.1109/TIT.2023.3274152](#).
- [11] **Ganghua Wang**, Jie Ding, and Yuhong Yang. “Regression with Set-Valued Categorical Predictors”. In: **Statistica Sinica** 33.4 (2023), pp. 2545–2560. DOI: [10.5705/ss.202021.0332](#).
- [12] Xun Xian, **Ganghua Wang**, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “A Unified Framework for Inference-Stage Backdoor Defenses”. In: **Proc. NeurIPS** (2023). [\[pdf\]](#).
- [13] Xun Xian*, **Ganghua Wang***, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Understanding backdoor attacks through the adaptability hypothesis”. In: **Proc. ICML** (2023). [\[pdf\]](#).

Under review

- [14] **Ganghua Wang**, Zhaorun Chen, Bo Li, and Haifeng Xu. “Cer-Eval: Certifiable and Cost-Efficient Evaluation Framework for LLMs”. In: **Proc. ICML** (2026).
- [15] **Ganghua Wang**, Oliver R. Wang, Bo Li, and Haifeng Xu. “Extrapolating Large Models from the Small: Optimal Learning of Scaling Laws”. In: **Proc. ICML** (2026).
- [16] Yangjianchen Xu, Qinglong Tian, and **Ganghua Wang**. “Semiparametric Regression Analysis of Right-Censored Events with Privacy-Preserving Data”. In: **Biometrics** (2026).
- [17] **Ganghua Wang** and Jie Ding. “Subset Privacy: Draw from an Obfuscated Urn”. In: **arXiv preprint** (2025). [\[pdf\]](#).
- [18] Wenjing Yang*, **Ganghua Wang***, Jie Ding, and Yuhong Yang. “A Theoretical Understanding of Neural Network Compression from Sparse Linear Approximation”. In: **IEEE Trans. Signal Process.** (2025). [\[pdf\]](#).

Manuscript

- [19] **Ganghua Wang**, Yuhong Yang, and Jie Ding. “Regression with Adversarially Perturbed Responses”. In: **Preparation** (2026).
- [20] **Ganghua Wang**, Zhiyuan Tang, and Jie Ding. “Classification with set-valued labels”. In: **Preparation** (2025).

HONORS AND AWARDS

Awarded by University of Chicago, Data Science Institute	
Faraco Postdoctoral Fellowship for Outstanding Research	2025
Mentor of the Year Award	2025
Awarded by University of Minnesota, Twin Cities	
Doctoral Dissertation Fellowship Finalist	2023
School of Statistics Travel Award	2023
Summer Research Fellowship	2020
School of Statistics First Year Scholarship	2019
Awarded by Cisco Systems, Inc.	
Cisco Research Graduate Award	2022
Cisco Research Fellow	2022, 2023
Awarded by Peking University	
The Academic Excellence Scholarship (3 times)	2016 - 2018

Fang Zheng Scholarship	2017
Wu Si Scholarship	2016, 2018
Freshman Scholarship	2015

TEACHING AND MENTORING

Instructor, University of Arizona <i>MATH 567A, Theoretical Statistics I</i>	<i>Fall 2026</i>
Mentor of Data Science Clinic, University of Chicago Mapping Human Rights Violations in the Palm Oil Industry <i>Inclusive Development International</i>	<i>Spring 2025</i>
Climate change-induced multi-hazard risks: Landslides <i>University of Rwanda</i>	<i>Winter 2024</i>
Characterizing African American Young Adult Novels' Narrative <i>University of Northern Iowa</i>	<i>Autumn 2024</i>
Mentor of Undergraduate Reading Group, School of Statistics, University of Minnesota	<i>2024</i>
AEOP High School Apprenticeship K-12 Outreach Program to enhance diversity, equity, and inclusion (sponsored by Army Research Office)	<i>2022</i>
Teaching Assistant, University of Minnesota, Twin Cities <i>STAT 4102 Theory of Statistics II</i>	<i>Fall 2020</i>
<i>STAT 3021 Introduction to Probability and Statistics</i>	<i>Spring 2020</i>
<i>STAT 3011 Introduction to Statistical Analysis</i>	<i>Fall 2019</i>

SERVICES

Organization Session at INFORMS annual meeting (co-organizing with Fan Yao), San Francisco, CA	<i>Nov. 2026</i>
Member Graduate Student Liaison Committee, School of Statistics, University of Minnesota	<i>June 2023 - Aug. 2024</i>
Review Journal of the American Statistical Association, IEEE Transactions on Information Theory, Transactions on Machine Learning Research, SIAM Journal on Mathematics of Data Science, Nature Electronics, AISTATS, AAAI, ICML, ICLR, ICASSP, NeurIPS, Neural Processing Letter, Machine Learning	

SOFTWARE

Python Packages

1. **SubsetPrivacy**: Implement the subset privacy mechanisms proposed in [17] and statistical inference methods for set-valued observations [11].
2. **ModelPrivacy**: Implement the common and proposed model stealing defense/attack strategies on benchmark datasets [2].
3. **FedGFT**: Implement a bias mitigation federated learning algorithm proposed in [6]. It has been incorporated into **Flame**, an open-source project for federated learning systems, which is developed by Cisco Systems, Inc.

4. **CBD**: Conformal backdoor defense that detects the backdoor inputs in the inference stage with provable guarantees on false positive rate [12].
5. **RAW**: A robust and agile watermarking technique for AI-generated images [8].

PATENTS

1. M. Lee, G. Wang, A. Payani, R. Kompella, “Group Bias Mitigation in Federated Learning Systems,” 07/28/2023, US patent #18/227,535

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, R, SQL

PRESENTATIONS AND TALKS

Joint Statistical Meeting, Boston, MA	Aug. 2026
ICSA Applied Statistics Symposium, Arlington, VA	June 2026
Department of Mathematics, University of Arizona, Tucson, AZ	Feb. 2026
Department of Decision Analytics and Operations, City University of Hong Kong, Hong Kong	Feb. 2026
Department of Statistics, University of Georgia, Athens, GA	Jan. 2026
Data Science Institute, University of Chicago, Chicago, IL	Dec. 2025
Guest Lecturer, Clemson University, Clemson, SC	Nov. 2025
Data Science Institute Research Day, University of Chicago, Chicago, IL	May 2025
Clemson University, Clemson, SC	Oct. 2024
International Conference on Econometrics and Statistics, Beijing, China	July 2024
International Conference on Learning Representations, Vienna, Austria	May 2024
Department of Statistics, Chinese University of Hong Kong, Hong Kong	Apr. 2024
Conference on Neural Information Processing Systems, New Orleans, LA	Dec. 2023
School of Statistics, University of Minnesota, Minneapolis, MN	Nov. 2023
Joint Statistical Meeting, Toronto, Canada	Aug. 2023
International Conference on Econometrics and Statistics, Tokyo, Japan	Aug. 2023
International Conference on Machine Learning, Honolulu, HI	July 2023
Joint Conference on Statistics and Data Science, Beijing, China	July 2023
Center of Statistical Research, Southwestern University of Finance and Economics, Chengdu, China	June 2023
Center for Statistical Science, Peking University, Beijing, China	June 2023
International Conference on Learning Representations, Kigali, Rwanda	May 2023
School of Statistics Student Seminar, Minneapolis, MN	Mar. 2023